

Document Control

- Document Title: Business Requirements Document (BRD)
- System Name: Application

Executive Summary

- This document outlines the business requirements for ingesting customer and order CSV data into Delta tables, cleansing and enriching data, executing SCD Type 2 handling for customer changes within an ordersummary table, and creating a daily customer spend aggregation. The implementation is to be done using Delta Live Tables as the data pipeline framework.
- The BRD is augmented with a complete Entity Relationship Diagram (ERD) for reference.

Business Objectives

- Load source CSVs into Delta tables for customer and order data.
- Cleanse customer and order data by removing "Null"/Null literals and duplicates.
- Enrich orders with TotalAmount ($\text{PricePerUnit} \times \text{Qty}$).
- Create ordersummary by joining customer and order tables, applying SCD Type 2 for customer changes.
- Maintain historical changes of customer data via SCD Type 2 handling.
- Generate customeraggregatespend table with aggregated spend per customer per date.
- Implement end-to-end pipeline using Delta Live Tables.

In-Scope

- Source ingestion from specified file system locations.
- Creation and management of Delta tables: customer, order, ordersummary, customeraggregatespend.
- Removal of "Null"/Null literals and duplicates.
- Computation of TotalAmount in the order table.
- Join logic and SCD Type 2 handling for customer dimension in ordersummary.
- Aggregation from ordersummary to customeraggregatespend.
- Use of Delta Live Tables for implementation.

Out-of-Scope

- Data sources, systems, or fields not present in source JSON.
- Security, access controls, SLAs, monitoring, and alerting.
- Performance tuning and scaling parameters.
- Scheduling, orchestration, or DevOps details not listed in requirements.
- Data quality metrics beyond specified cleansing.

- User interfaces, reports, or analytics layers.

Stakeholders

- Information not present in source JSON.

Business Requirements

BR-APP-001 — Source Ingestion

- Description: Load source CSVs into Delta tables located at specified file system paths for customer and order data.
- Business Value: Enables automated data ingestion from external sources for analytics and reporting.
- Priority: High

BR-APP-002 — Schema Definition

- Description: Define the customer table with columns CustId, Name, EmailId, Region, and order table with columns OrderId, ItemName, PricePerUnit, Qty, Date, CustId.
- Business Value: Ensures data consistency and supports downstream data processing.
- Priority: High

BR-APP-003 — Computed TotalAmount Column

- Description: Compute TotalAmount in the order table as $\text{PricePerUnit} \times \text{Qty}$.
- Business Value: Supports spend analytics and customer value calculation.
- Priority: High

BR-APP-004 — Data Cleansing

- Description: Remove all "Null"/Null literal values and duplicate records from customer and order tables.
- Business Value: Maintains data quality and reliability for decision-making.
- Priority: High

BR-APP-005 — Ordersummary Table Creation

- Description: Create the ordersummary Delta table, populated by joining customer and order on CustId.
- Business Value: Centralizes enriched order data for downstream consumption.
- Priority: High

BR-APP-006 — SCD Type 2 Handling

- Description: On ordersummary, implement SCD Type 2 change logic for customer dimension, maintaining historical versions and updating status fields.
- Business Value: Ensures auditability and accurate reporting on customer changes over time.
- Priority: High

BR-APP-007 — Aggregated Customer Spend

- Description: Create customeraggregatespend table aggregating TotalAmount by Name and Date, sourced from ordersummary.
- Business Value: Enables analysis on customer spend trends and patterns.
- Priority: High

BR-APP-008 — Delta Live Tables Implementation

- Description: Use Delta Live Tables to orchestrate the full ETL pipeline as per specified requirements, ensuring dependencies and ordering.
- Business Value: Enables scalable, maintainable, and automated ETL processing.
- Priority: High

Assumptions & Constraints

- No requirements beyond those in the provided source JSON will be supported.
- Delta Live Tables is mandated as the implementation approach.
- SCD Type 2 management references business rules but does not explicitly define all tracking fields.

Success Metrics

- 100% of customer and order records accurately ingested with specified schema.
- All duplicates and "Null"/Null literals are eliminated prior to downstream transformation.
- Ordersummary accurately reflects SCD Type 2 logic for customer changes.
- customeraggregatespend contains accurate daily sums of TotalAmount by Name.
- All pipeline steps orchestrated successfully in Delta Live Tables with no errors.

Risks & Mitigations

- Missing explicit field definitions for SCD Type 2 tracking: Review and clarify with data owners.
- Data quality may not be addressed beyond removal of "Null"/Null literals and duplicates: Plan for enhancement.
- Lack of security, monitoring, and alerting: Identify in future detailed design phases.

Approval

- Approvers: To be confirmed by project sponsor.

- Approval Date: To be determined.

Appendix

Appendix A: Source Definitions

Name	Type	Location	Format	Notes
Customer source	CSV	/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/customerdata	Not present in source JSON	Loaded into Delta customer table
Order source	CSV	/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/orderdata	Not present in source JSON	Loaded into Delta orders table

Appendix B: Target Definitions

	Table	Columns	Notes
Not present in source JSON	customer	CustId, Name, EmailId, Region	Loaded from source
Not present in source JSON	order	OrderId, ItemName, PricePerUnit, Qty, Date, CustId	Loaded from source
	ordersummary	CustId, Name, EmailId, Region, OrderId, ItemName, PricePerUnit, Qty, Date	Join of customer and order
	customeraggregatespend	Name, TotalAmount, Date	Aggregation of ordersummary

Appendix C: Transformations

Step	Transformation	Input	Output	Logic
1	Data Cleansing	customer, order	cleansed data	Remove Nulls, deduplicate
2	Compute column	order	order	TotalAmount = PricePerUnit × Qty
3	Join	customer + order	ordersummary	On CustId
4	SCD Type 2	ordersummary	ordersummary (historical)	Update status/history
5	Aggregate	ordersummary	customeraggregatespend	Sum(TotalAmount), group by Name, Date

Appendix D: Business Rules

Rule ID	Area	Description	Source
BR-1	Computed Column	TotalAmount = PricePerUnit × Qty	Requirement
BR-2	Cleansing	Remove Null/Null and duplicates	Requirement
BR-3	Join	customer to order on CustId	Requirement
BR-4	SCD Type 2	Maintain full history, set fields accordingly	Requirement
BR-5	Aggregation	Sum TotalAmount by Name, Date	Requirement

Appendix E: Process Flow

Step	Description	Dependency
1	Load customer and order data from CSV to Delta	Source path availability
2	Cleanse loaded Delta tables	Successful loading

3	Compute TotalAmount for orders	Cleansed order data
4	Create ordersummary by joining customer/order and apply SCD Type 2 logic	Cleansed data, computed column
5	Aggregate ordersummary to create customeraggregatespend	ordersummary table

Appendix F: Entity Relationship Diagram (ERD)

- The following ERD represents the relationship between the core entities.
- customer (CustId as PK)
- order (OrderId as PK, references customer.CustId)
- ordersummary (CustId, Name, EmailId, Region, OrderId, ItemName, PricePerUnit, Qty, Date; combines/joined from customer and order for SCD Type 2 management)
- customeraggregatespend (Name, TotalAmount, Date; aggregated from ordersummary)

Entity Relationships:

- customer 1--* order (linked by CustId)
- customer 1--* ordersummary (linked by CustId)
- order 1--* ordersummary (linked by OrderId)
- ordersummary *--* customeraggregatespend (linked by Name, Date)

See below for ERD structure:

[customer]

*CustId

Name

EmailId

Region

[order]

*OrderId

ItemName

PricePerUnit

Qty

Date

CustId

[ordersummary]

CustId

Name

EmailId

Region

OrderId

ItemName

PricePerUnit

Qty

Date

[customeraggregatespend]

Name

TotalAmount

Date

customer 1--* order

customer 1--* ordersummary

order 1--* ordersummary

ordersummary *--* customeraggregatespend

(For a visual ERD, refer to the generated ERD diagram.)

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